

2. Take the action allotted to the identified key

The key identification is done by identifying the row and the column to which the key belongs. Fig. 1 shows how the keypad is connected to Port B of the MCU. The lower nibble of Port B is declared as output pins (scan lines) and the upper nibble is declared as input pins (return lines). The number 0Fh is written to Port B so that the lower four bits become high and the upper four bits become low.

Whenever a key is pressed, the upper nibble pin (return line) of Port B, to which the column containing the key is connected, goes high. Thus the column is identified. Column identification is done using a switch-case block in the main program. On identifying the column, the rowfind(int) function is called, which does the job of row identification. To identify the row, scan lines are made low one by one in sequence and status of the return line corresponding to the key is checked. If it becomes low, the key belongs to that scan line or row. The row and column numbers are stored in two global variables 'row' and 'col.' A key debouncing delay of 20 ms is provided in the program by calling the DelayMs(20) function.

After identifying the key, the action() function is called in the main program, to perform the action corresponding to the identified key.

The detailed procedure for developing the project using MPLAB IDE, compiling the same using Hitech C compiler and burning the executable hex file to the microcontroller was explained in 'Construction' section of

EFY's May issue.

The above description is available in a file named 'lock.c.' Functions lcd_init(), lcd_goto(int), lcd_clear() and lcd_putch(char) are defined in a file named 'lcd.c' and the DelayMs(int) function is defined in the delay.c file. Add all the three 'C' files—lock.c, lcd.c and delay.c—as source files to the MPLAB IDE project. Save the project file as 'Elock.mcp.' Set configuration bits properly before building the project. Select the oscillator as XT and disable all other features like watchdog timer, power-up timer and brownout detection. After successfully building the project, the Elock.hex file is generated. Burn it into the chip using a suitable programmer, e.g., MPLAB ICD2.

Testing

Once the program is burnt into the chip and the hardware setup is ready, the user can test the system. When the power supply is switched on, message "Welcome" is displayed on the LCD screen. The default password set in the program is 1234.

To turn on the appliance, press 'Open' key. The system will ask for the password. Enter the password as 1234 and press 'Go.' The appliance should turn on (RC2=1) and the message "Password Accepted" should be displayed for two seconds followed by the message "Lock Open."

To turn off the appliance, press 'Lock.' The appliance should immediately turn off (RC2=0) and the message "Lock Closed" should be displayed on the LCD screen.

To set a new password, press 'New'

key. The system should ask for the current and new passwords. Press 'Go' each time after you enter the four-digit password. The message "Password Saved" should appear for two seconds, followed by the message "Welcome." Now you can turn on the device by pressing 'Open' and then entering the new password that you have set. Key 'C' acts like 'Backspace' key in a PC's keyboard.

The only disadvantage of this system is that the password set by the user is stored in the RAM and hence it is lost when the supply is switched off. When the supply is switched on again, the system is reset with the default password 1234.

Construction

A single-side, solder-side PCB layout for the PIC microcontroller-based electronic lock is shown in Fig. 4 and its component layout in Fig. 5.

Connect the 4×4 matrix keypad and 16×2 LCD to PIC16F877A microcontroller as shown in Fig. 1. Complete the remaining connections also as shown in Fig. 1. Connect the appliance to be controlled to RC2 (pin 17) of Port C through the relay-driver circuit as shown in Fig. 3. Connect 6V power supply from the power supply circuit to the relay driver circuit. The 5V supply required by the microcontroller is obtained from the 7805 regulator output.

EFY note. The complete project folder (named Elock.zip) containing the source code and related files is included in this month's EFY-CD and also available on www.efymag.com website. ●